GEM Alpine Business School

GROUP PROJECT · 2026-2027

Project Guidelines

Fundamentals of Artificial Intelligence and Machine Learning
MSc Managing with Data & Artificial Intelligence M2

September 2026-2027

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This project develops your digital fluency and analytical judgement through an applied AI study of a business problem. In groups, you design an end-to-end AI solution — from data to a KPI-measured business decision — and pitch it in **3 minutes** at the last session. It follows the course's five steps: four datasets, preparation and enrichment, the core AI engines (with optional Python extensions), exploiting the results, and storytelling.

The 3-minute pitch is only an early, rapid presentation of a project that is **not yet finished**. The finished work is handed in two weeks later, as an archive that includes a **detailed analysis report (PDF)**: the full write-up of your project, with explanations, graphs, metrics and interpretation. See **The detailed analysis report** below.

1. Description

In groups of 2 or 3, you take a business problem and design an end-to-end AI solution. It must be **doable with Orange Data Mining alone**: use the three core engines Orange supports — Machine Learning, Deep Learning and Natural Language Processing — and optionally add Python extensions (Reinforcement Learning, CNN/RNN/LSTM).

Everything must be defensible in a professional setting. Take an explicit role (e.g. data & AI consultant for company X, studying market Y) and justify your choices against it; without a role, state case by case whom each model, indicator or chart helps, and when. Relevance and usefulness are not the same: an element that adds nothing today earns its place if you say when it would ("if this market shifts, this feature quantifies…"). For every element, you must be able to say whom it serves and when.

Follow the value chain end to end: Problem → Data → Prepare & enrich → Build AI solutions → Present results. A strong project is not the one with the most tools, but the one with the clearest business story and a measured value.

Organisation

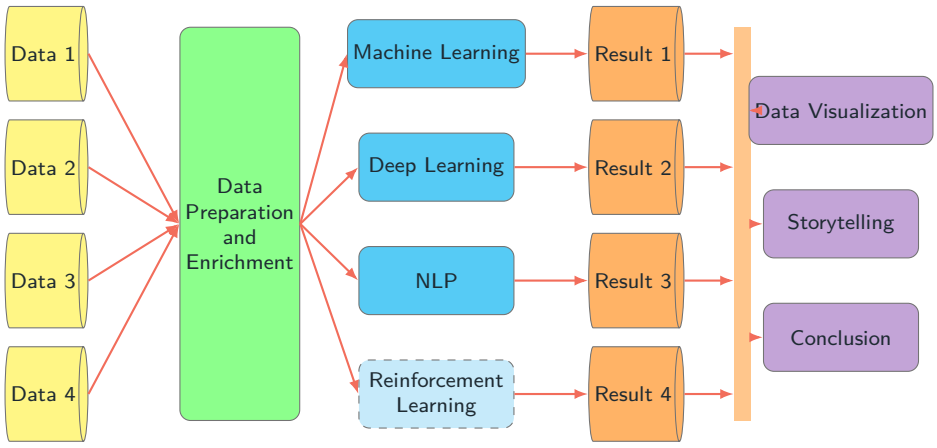
Group work is **not part of the sessions**: you organise the project yourselves, outside class time. You can, however, use the sessions to ask the professor for advice and feedback. Each session advances a step of your project.

Assessment

The collective assessment counts for **40% of the final grade** (the other 60% is the individual Moodle quiz). It combines the group project and a 3-minute pitch, graded on substance and delivery. See **Deadlines** below.

2. How the project is built — the 5 steps

The project follows the structure seen in class, from four datasets to a business story. Follow the five steps in order.



The project at a glance: four datasets, prepared and enriched, feed the AI engines whose results become a business story. Solid border = works in Orange; dashed = optional, needs Python.

2.1 Step 1/5 — Data sources

- Identify **four publicly available datasets** that can help you conduct your project.
- Datasets can be related, add value to the final conclusion, be independent, or be a source of information you created yourself.
- They must be public and must not disclose personal or confidential information.
- Data must be both **quantitative and qualitative**.
- More than **1000 records** and more than **25 features of interest** is better.

2.2 Step 2/5 — Data preparation and enrichment

Preparing data is about:

- identifying and managing extreme values;
- taking care of missing values;
- converting categorical features to numeric;
- converting text when needed;
- computing aggregate features;
- standardizing values for computation.

Enriching data is about identifying additional data to build more value in the dataset: bank holidays, weather, stock prices, market insights or other.

2.3 Step 3/5 — The AI engine

The project must be **doable with Orange Data Mining alone**. The three core engines are required and fully supported by Orange:

- a **Machine Learning** algorithm of supervised and/or unsupervised learning;
- a **Deep Learning** network — the Orange *Neural Network* widget (a feed-forward network with hidden layers);
- at least one **Natural Language Processing** algorithm — Orange *Text Mining* add-on (pre-process text, bag of words, sentiment, topic modelling…).

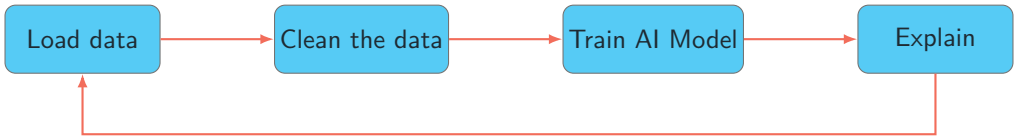
Two further aspects of the course are **optional and rewarded**, because they require Python:

- advanced **Deep Learning** (CNN, RNN, LSTM) — for images and sequences;
- one **Reinforcement Learning** exercise (customer recommendation or stock trading).

So the same project can be delivered at two levels: *Orange-only* (the three core engines) or *Orange + Python* (add the optional engines). Python is never required, but it is valued.

> **Orange-only vs Python**
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> **Orange-only — sufficient:** Machine Learning, Deep Learning (feed-forward network), NLP via the Text Mining add-on.
> **Python — optional, rewarded:** CNN / RNN / LSTM, Reinforcement Learning, transformers / LLM.

You may implement the engines on a non-business dataset or topic, although a business framing is advised. The classic workflow for each engine is a loop, not a straight line:



Load data → clean the data → train the AI model → explain — and iterate.

2.4 Step 4/5 — Exploiting the results

AI helps us understand, enrich and convert data. For each of your analyses, save the outputs somewhere — csv, Excel, SQL, NoSQL, whatever you like. These data let you compare against real-life future data, and they serve to tell a story.

2.5 Step 5/5 — Telling a story

Once new data has been generated it must serve a business purpose. Express a story based on clear visuals, support your conclusion with clear metrics, conclude on the value of using AI on your project, and make sure the conclusion comes with clear data and depicts an added value from AI.

3. Examples of possible subjects

To inspire you, here are subject ideas — all doable with Orange:

- Customer segmentation — unsupervised Machine Learning (k-means);
- Sentiment analysis on the Enron e-mails — NLP;
- Customer churn prediction — classification;
- Bundesliga ticket-price prediction — regression;

- Retail sales forecasting — regression;
- Employee-retention strategies and criteria — classification;
- Topic modelling of customer reviews — NLP;
- Sports bets — classification;
- Performance in Formula 1 — regression;
- Fraud detection — Random Forest classification.

3.1 Worked examples

Each example below gives a concrete path — sources, analyses, expected results and the business conclusion — and stays within what the course covers and Orange can do. They are starting points: find your own data and justify your choices.

Customer segmentation — unsupervised Machine Learning

Sources: a public retail dataset (e.g. UCI *Online Retail*, *Mall Customers*) with purchase history and demographics.

Analyses: clean the data, standardize the numeric features (Orange *Preprocess*), run **k-means** (Orange *k-Means*), then read the segments in a *Scatter Plot* and the *Data Table*.

Results: 3–5 segments (e.g. high-value, occasional, at-risk customers) with their profiles.

Conclusion: target each segment with a tailored campaign → a measurable uplift in retention or margin.

Fraud detection — supervised classification

Sources: a public credit-card fraud dataset (e.g. Kaggle *creditcard.csv*, ∼284k transactions, strongly imbalanced).

Analyses: check the class imbalance (Orange *Distributions*), train a **Tree / Random Forest** and evaluate in *Test & Score* with **precision, recall, F1 and ROC-AUC** rather than accuracy.

Results: AUC ≈ 0.97; the precision/recall trade-off and the *Confusion Matrix* expose the cost of a missed fraud versus a false alarm.

Conclusion: set the decision threshold from the cost of the two errors → fewer losses (cash-flow KPI).

Sentiment analysis on the Enron e-mails — NLP

Sources: the public *Enron* e-mail corpus (∼500k messages).

Analyses: with the Orange *Text Mining* add-on: *Preprocess Text*, *Bag of Words*, *Sentiment Analysis* and *Topic Modelling*; aggregate by sender and over time.

Results: sentiment distribution over time and by sender, word clouds, top topics.

Conclusion: use sentiment as an early-warning signal (churn, misconduct) → a monitoring decision.

Ticket-price prediction — regression

Sources: public match/attendance and pricing data plus team standings and fixtures (e.g. Bundesliga).

Analyses: engineer features (opponent rank, derby, day/time), compare *Linear Regression* and *Random Forest*; evaluate with **MSE / $\mathbf{R^2}$** in *Test & Score*.

Results: $R^2 \approx 0.7$ and the most influential features.

Conclusion: dynamic pricing / revenue optimization → a margin KPI.

House-price prediction — Deep Learning (Orange Neural Network)

Sources: a public housing dataset (e.g. Kaggle *House Prices*, *California Housing*).

Analyses: clean and standardize the features, train the Orange **Neural Network** (feed-forward), and compare it with *Linear Regression*; evaluate with **MSE / R^2**.

Results: R^2 and the error, plus whether the network beats the linear model.

Conclusion: use Deep Learning when the price drivers are non-linear → better pricing decisions.

Employee retention — classification

Sources: a public HR dataset (e.g. IBM *HR Analytics*, ∼1,470 employees).

Analyses: read the drivers with *Feature Statistics* and *Box Plot*, train *Logistic Regression* / *Random Forest* (cross-validation in *Test & Score*); evaluate with **ROC-AUC**.

Results: the main drivers (overtime, satisfaction, tenure) and the *Confusion Matrix*.

Conclusion: target retention actions on high-risk employees → a lower turnover cost.

4. Required elements

4.1 The solution

- Cover the three core engines with Orange (Machine Learning, Deep Learning, NLP), each with at least one concrete result. Depth matters more than quantity: one solid application per engine beats many rushed ones. The optional Python extensions (RL, CNN/RNN/LSTM) are rewarded but never required.
- Build it with Orange Data Mining (free, no-code) — required — and, optionally, Python for the advanced extensions. The choice of acquisition, cleaning and processing tools is otherwise free.
- The project must open and run on a machine other than yours: include your Orange workflows (.ows) and the data.
- A purely static study is not enough: the value chain must be complete, ending in a business decision and a measured KPI.

4.2 The data

- Public datasets only: four sources, with more than 1000 records and more than 25 features of interest, mixing quantitative and qualitative variables. Finding them is up to you.
- Company data: only if public. Internal, confidential or unpublished data are forbidden, even if you have access to them via an internship or work-study contract.
- Data extracted from an existing document by an AI tool: allowed if the acquisition and verification methods are detailed in the presentation.
- Fake / invented data, or personally identifying data: forbidden.

4.3 The project deck

Your project deck (part of the submission, and the basis of the pitch) must contain the following elements:

- the full names of all group members;
- an introduction: your data, their origin, the variables of interest, the role you take on if any, and the business question you address;
- the AI engines you chose and why (justify against the alternatives);

- your value chain: data, preparation and enrichment, models, and the business decision;
- the results, with honest metrics (accuracy, F1, AUC, confusion matrix…) and an interpretation of each visual;
- the business value, measured with KPIs (Sales, Margin, Cashflow);
- optional but valued: a walkthrough of your solution's features, and highlighting efforts that did not pan out if they are instructive (e.g. a Python script that failed);
- optional but valued: an AI-use statement, detailing what AI produced and how you checked that part of the work. The seriousness and honesty of this statement are graded.

For every indicator or visual, justify its relevance — in light of the chosen role, or by stating to whom and when it would be useful.

4.4 The detailed analysis report (PDF)

The 3-minute pitch is a quick teaser of a project still in progress. The finished deliverable is a **detailed analysis report**, submitted as a PDF with the final archive. Where the deck shows, the report *explains*: it is a written document you could hand to a manager or a client. It must contain:

- the business problem, the role you take on, and the question you address;
- the four data sources, why you chose them, and their preparation and enrichment;
- your methodology: the AI engines, the features, the settings, and the alternatives you rejected and why;
- the results, with clear graphs, tables and honest metrics, and a written interpretation of each visual;
- the business value, measured with KPIs (Sales, Margin, Cashflow);
- the limitations of your work and the AI-use statement;
- a conclusion and the list of sources / references.

Expected length: **about 15 to 20 pages** (excluding the cover page, table of contents and appendices), figures and tables included. Too short reads as superficial; padded with screenshots reads as filler. Depth, clarity and honest interpretation matter more than length.

5. Submission

Submit on Moodle, in your group's space, as a single upload (`project.zip`). **Deadline: 2 November 2026** (two weeks after the last class). The **3-minute pitch** takes place during the last session, on **19 October 2026**, and is graded on substance and delivery. The archive submitted two weeks later must contain the **detailed analysis report (PDF)** — the finished, written version of the project.

File	Contents	Required
project.zip	Single archive containing all items below.	Yes
report.pdf	Detailed analysis report — the written deliverable (see above).	Yes
slides.pptx	The deck you present.	Yes
workflows (.ows)	Your Orange Data Mining workflows.	Yes
The data	The data used and their sources.	Yes
notebook / code	Optional Python notebook or script.	No
read_me.txt	Helpful notes.	No

6. Grading

The collective assessment counts for 40% of the final grade (the other 60% is the individual Moodle quiz). It combines the group project and the 3-minute pitch, and is graded out of 20, shared by the group (except in cases of substantial and blatant unevenness of involvement).

Analytical judgement comes before everything else. Next come the command of your approach and the overall quality of your work. Technicality is not a criterion in itself. Indicative benchmarks:

- the three core engines covered with Orange, with simple but relevant solutions: 12 to 14;
- the three core engines used in depth, plus optional Python work (RL, CNN/RNN/LSTM), a dynamic or automated solution, and a convincing business story with KPIs: 14 and above;
- below 12: incomplete deliverable, unopenable file, unjustified choices, missing or superficial analysis.

7. Deadlines

- **3-minute pitch**: last session, **Monday 19 October 2026** — a quick, early presentation of the project in progress; you sell the idea and the direction.
- **Final project** (`project.zip`): submitted on Moodle by **2 November 2026**, two weeks after the last class — the complete work, including the **detailed analysis report (PDF)**.

8. Pitch structure (3 minutes)

Three minutes go fast: aim for **4–5 slides**, rehearse out loud, and *sell* your project.

Part	Time	Content
Hook & problem	30 s	The business question, who benefits, why AI.
Solution & data	60 s	The value chain and the engines used.
Results & value	60 s	One key metric and the KPIs (Sales / Margin / Cashflow).
Close	30 s	The impact, and what you would do next.

> **What the jury looks for**
>
> A clear business problem and justified AI engines.
> Honest metrics (accuracy, F1, AUC, confusion matrix).
> Measurable KPIs (Sales, Margin, Cashflow).
> Rehearse out loud — timing is part of the grade.